

```
import pandas as pd
import numpy as np
import cv2
import matplotlib.pyplot as plt
```

```
from google.colab import files
upload_cat = files.upload()
upload_dog = files.upload()
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving cat.4003.jpg to cat.4003.jpg
 No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun

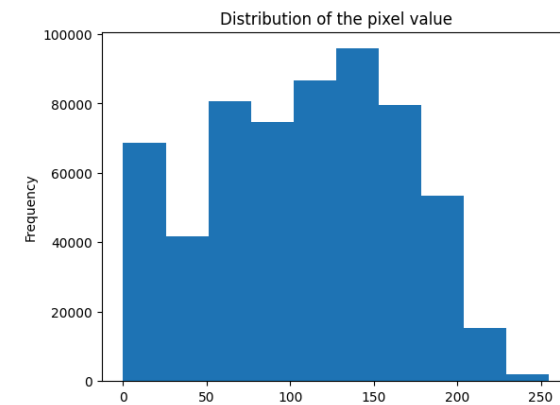
```
from PIL import Image
```

```
img_mpl = Image.open('cat.4003.jpg')
plt.imshow(img_mpl)
plt.show()
```

```
#img_mpl = plt.imread('cat.4003.jpg')
#img_cv2 = cv2.imread('cat.4003.jpg')
#img_mpl.shape, img_cv2.shape
```



```
pd.Series(img_cv2.flatten()).plot(kind='hist',
                                  bins=10,
                                  title='Distribution of the pixel value')
plt.show()
```



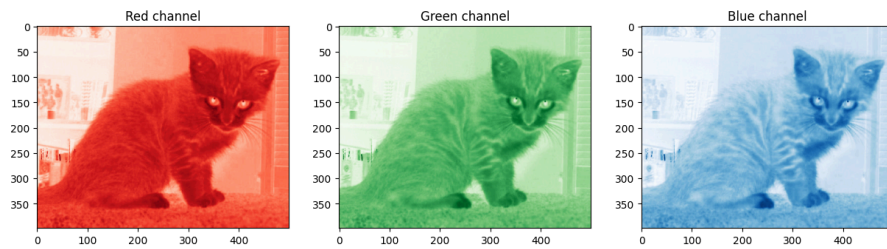
```
# Plotting a figure of width 10 and height 10
fig, ax = plt.subplots(figsize=(10, 10))
ax.imshow(img_mpl)
ax.axis('off')
plt.show()
```



```
# Display RGB Channels of our image
fig, axes = plt.subplots(1, 3, figsize=(15, 10))

axes[0].imshow(img_mpl[:, :, 0], cmap='Reds')
axes[1].imshow(img_mpl[:, :, 1], cmap='Greens')
axes[2].imshow(img_mpl[:, :, 2], cmap='Blues')

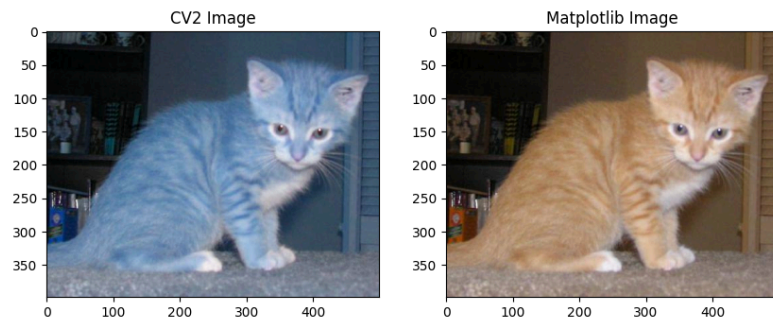
axes[0].set_title('Red channel')
axes[1].set_title('Green channel')
axes[2].set_title('Blue channel')
plt.show()
```



```
# cv2 reads in channels as BGR
# matplotlib reads in channels as RGB
fig, axs = plt.subplots(1, 2, figsize=(10, 5))
```

```
axs[0].imshow(img_cv2)
axs[1].imshow(img_mpl)

axs[0].set_title('CV2 Image')
axs[1].set_title('Matplotlib Image')
plt.show()
```



```
# Converting from BGR to RGB
img_cv2_rgb = cv2.cvtColor(img_cv2, cv2.COLOR_BGR2RGB)
fig, ax = plt.subplots()
ax.imshow(img_cv2_rgb)
ax.axis('off')
plt.show()
```



```
# Image Manipulation
img = plt.imread('dog.4817.jpg')
```

```
fig, ax = plt.subplots(figsize=(8, 8))
ax.imshow(img)
ax.axis('off')
plt.show()
```



```
img_gray = cv2.cvtColor(img, cv2.COLOR_RGB2GRAY)
fig, ax = plt.subplots(figsize=(8, 8))
ax.imshow(img_gray, cmap='Greys')
ax.axis('off')
ax.set_title('Grey Image')
plt.show()
```

Grey Image



```
# Resizing and Scaling
img_resized = cv2.resize(img, None, fx=0.3, fy=0.3)
fig, ax = plt.subplots(figsize=(8, 8))
ax.imshow(img_resized)
ax.axis('off')
plt.show()
```



```
img_resize = cv2.resize(img, (5000, 5000), interpolation = cv2.INTER_CUBIC)
print(img_resize.shape)
fig, ax = plt.subplots(figsize=(8, 8))
ax.imshow(img_resize)
ax.axis('off')
plt.show()
```

(5000, 5000, 3)

